FlowGrid

SCAN ME

Adaptive Traffic Light Control System

Real-Time Intersection Optimization Empowered by Computer Vision & Deep Reinforcement Learning

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Background

The Challenge

Urban intersections are plagued by inefficient signal timing, leading to congestion, increased wait times, and environmental impact. Fixed-time controllers fail to adapt to stochastic traffic flow, often causing bottlenecks even in low-density scenarios.

Our Idea

An intelligent agent that treats the intersection as a dynamic system. By replacing rigid timers with a learned policy, we enable the intersection to "respond" to traffic pressure and vehicle wait-time in real-time, optimizing flow and reducing frustration.

Key Features (FR)

- **Safety-Constrained Execution:** Native SUMO integration ensures phase safety with yellow light transitions and minimum green enforcement.
- **Dynamic Intelligence:** A 21-dimensional state space that captures real time demand across all lanes.
- **Starvation Prevention:** Explicit "starvation" metrics ensure no vehicle waits indefinitely, effectively handling edge cases.
- **Adaptive Fairness:** A cyclic but flexible phase sequence that balances flow throughput with individual lane fairness.

Solution & Technologies

The Intelligence Core (The Brain)

The heart of our system is an advanced Maskable Proximal Policy Optimization (PPO) agent.

- **On-Policy Stability:** The agent learns exclusively from fresh, current rollouts. This ensures that the policy updates are always aligned with the reality of the traffic conditions, preventing the "stale experience" problem found in off-policy models.
- **Clipped Surrogate Objective:** To ensure reliable behavior, we employ a clipped gradient mechanism (ε=0.2). This prevents the agent from making catastrophically large policy shifts, ensuring steady and predictable learning behavior.
- **Stochastic Policy:** Instead of a rigid "if-then" rule, the agent outputs a probability distribution over actions. This allows the agent to handle the inherent randomness of traffic flow and express uncertainty when traffic conditions are ambiguous.

Sumo Simulator Preview

Future Work

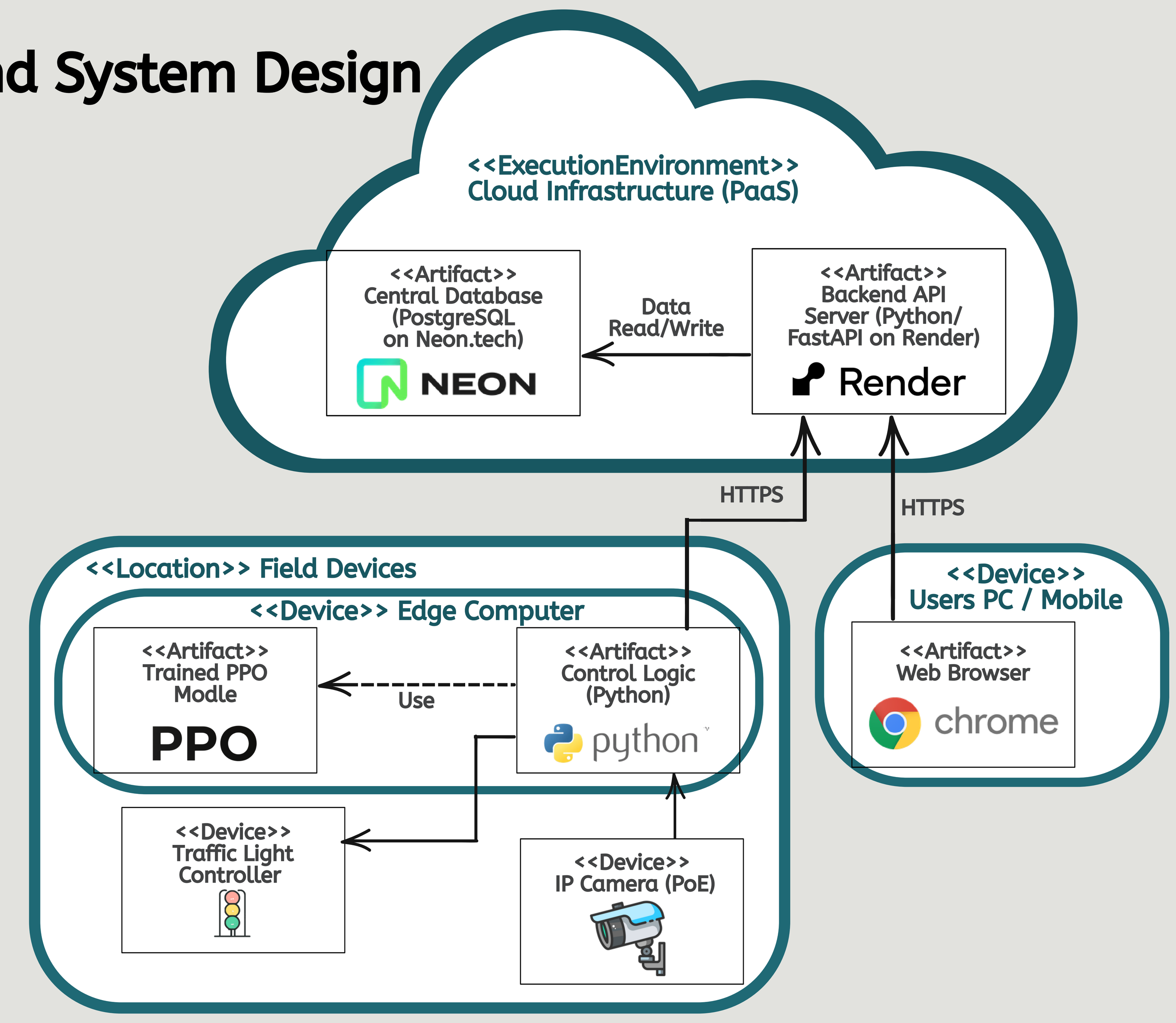
From Simulation to Real-World Sensing

Our current architecture relies on direct TraCI simulation data. To bridge the gap to real world infrastructure, our next development phase focuses on Computer Vision integration:

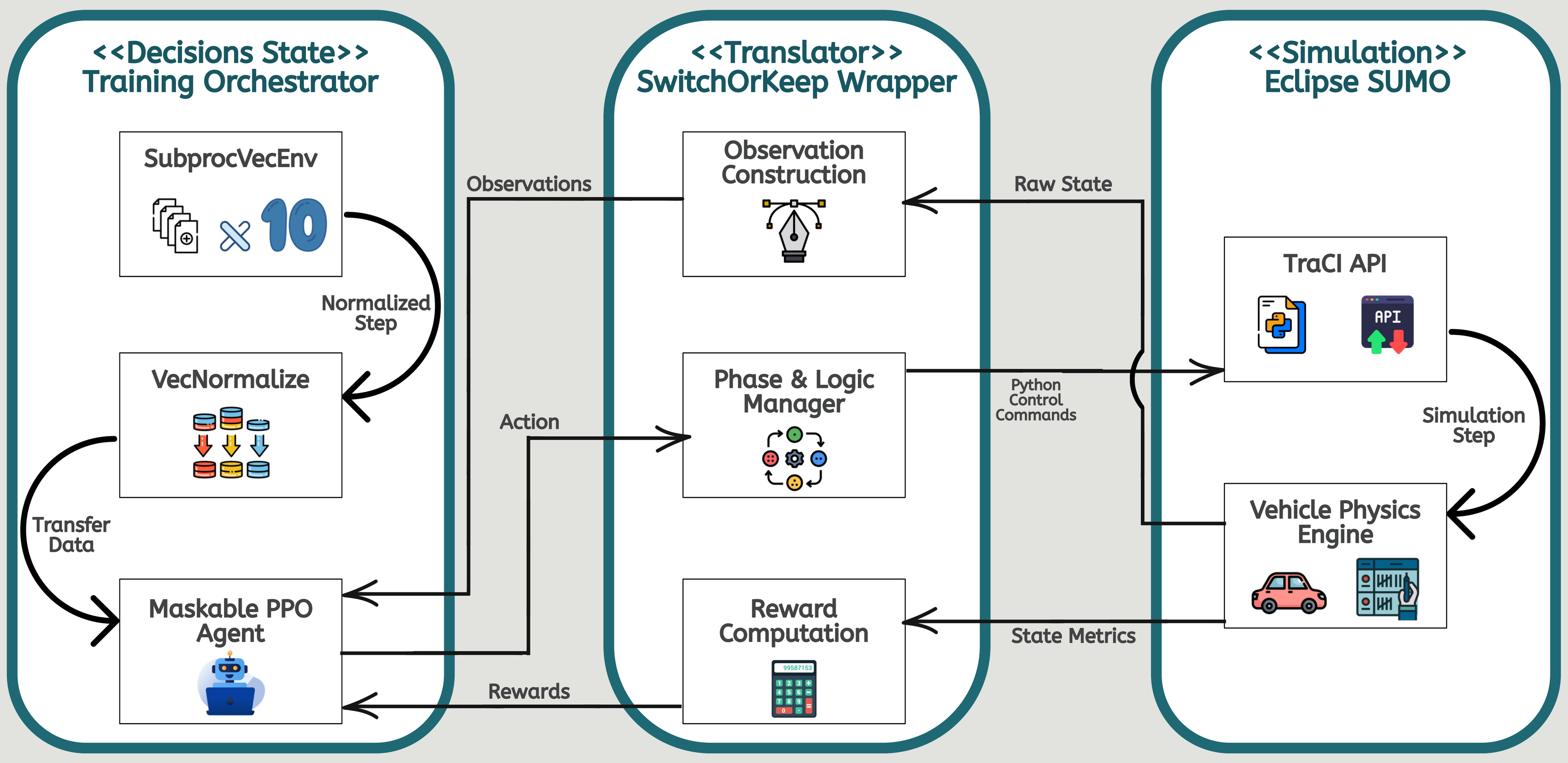
- **Object Detection:** Integrating YOLO to identify and classify vehicles in real-time from standard traffic CCTV feeds.
- **Object Tracking:** Utilizing DeepSORT to maintain consistent vehicle IDs across frames, enabling accurate demand estimation and queue length measurement.

Architecture

End-to-End System Design



Core Intelligence Framework



KPIs, Metrics & Results

Performance Benchmarks (Medium-Traffic Scenario)

The PPO agent and baseline controllers were tested against **identical** traffic arrival distributions to isolate algorithmic impact.

Performance: FlowGrid achieves near-optimal results compared to fixed-time baselines while maintaining full adaptive capabilities.